INTRODUCTION TO

Introduction of Electronics

Module 2 – Robotics and Automation Systems

PRESENTED BY

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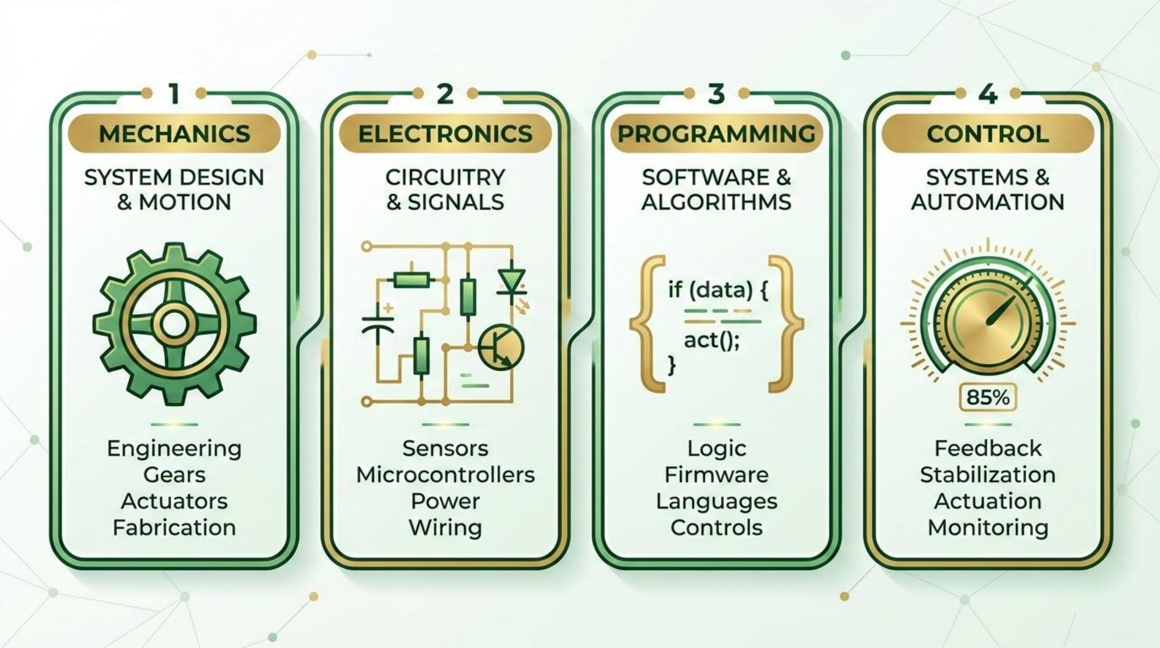
Module Overview

Key topics covered in electronics for robotics:

- What electronics means
- Why electronics is used in robotics
- Basic electricity concepts
- Voltage, resistance, and current
- Ohm's Law applications
- Series and parallel resistance

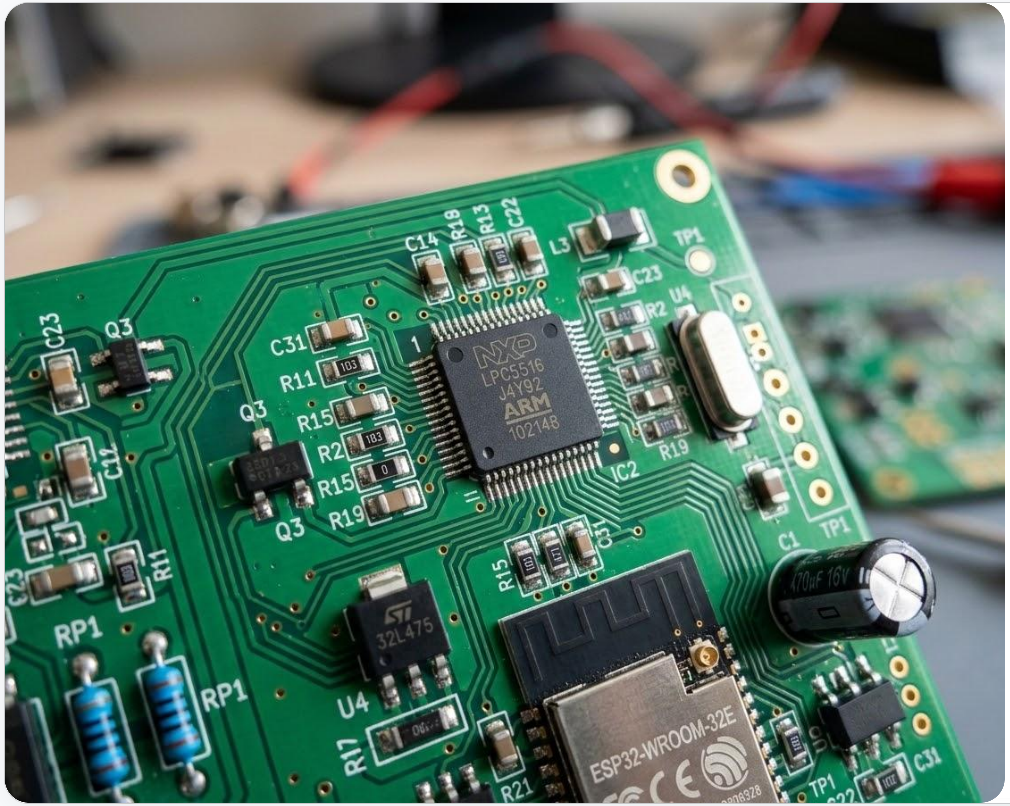


Learning Objectives

By the end of this module, students will learn to:

- Define electronics
- Explain electricity and its types
- Understand voltage and resistance
- Apply Ohm's Law
- Use the resistor color code

What is Electronics?



DEFINITION

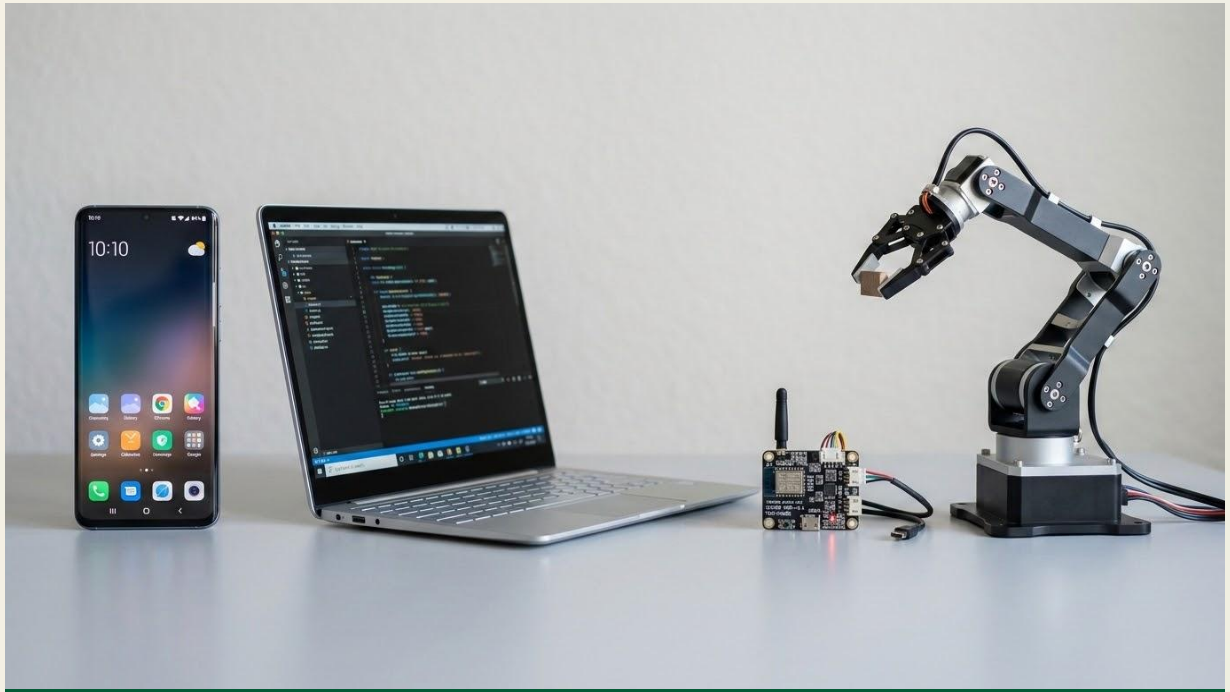
“Electronics is the study and use of electrical components to control the flow of current in circuits.”

Foundational concept for Robotics and Automation Systems, enabling precise control of robotic behavior.

Electronics in Daily Life

Everyday applications include:

- Mobile phones
- Televisions
- Computers
- Sensors
- Robots
- Automation machines



Integrated Electronics: From Mobile Code to Robotic Execution

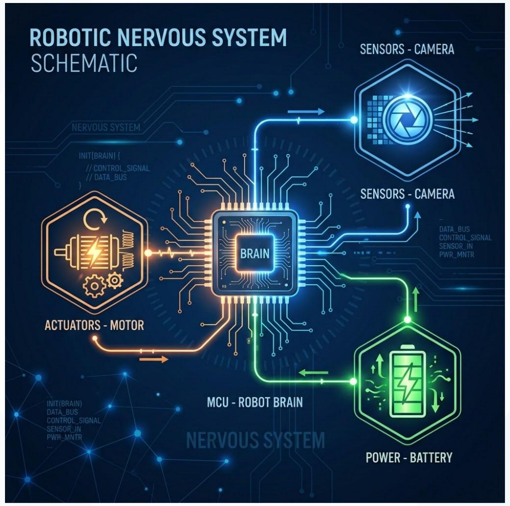
Why Electronics is Used in Robotics

Electronics form the **nervous system** of a robot:

Controls Robot Movement

Reads Sensor Data

Powers Motors



ROBOT BRAIN

Sends and Receives Signals

Makes Automation Possible

Electronics and Robotics Connection

INPUT

Sensors give input data from the environment.

PROCESS

The Controller processes the data.

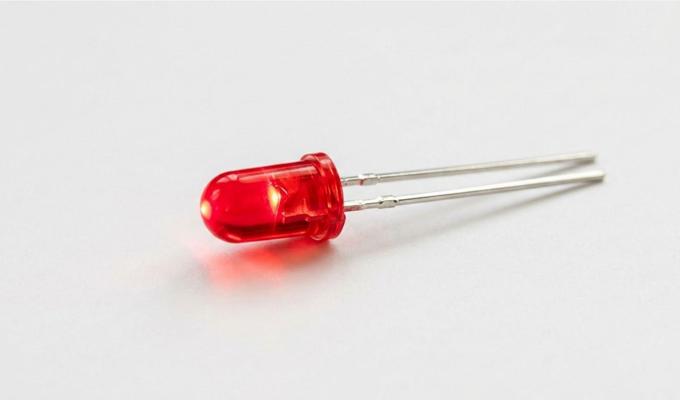
OUTPUT

Motors and actuators create the physical output.

Basic Electronic Components

Battery

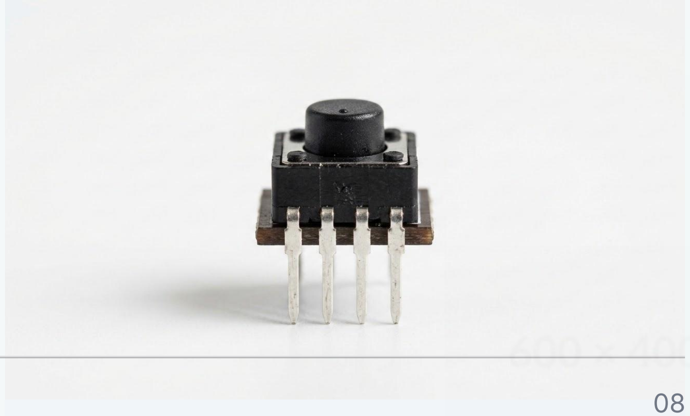
Resistor



LED

Motor

Sensor



Switch

What is Electricity?

DEFINITION

"Electricity is the movement or presence of electric charge."

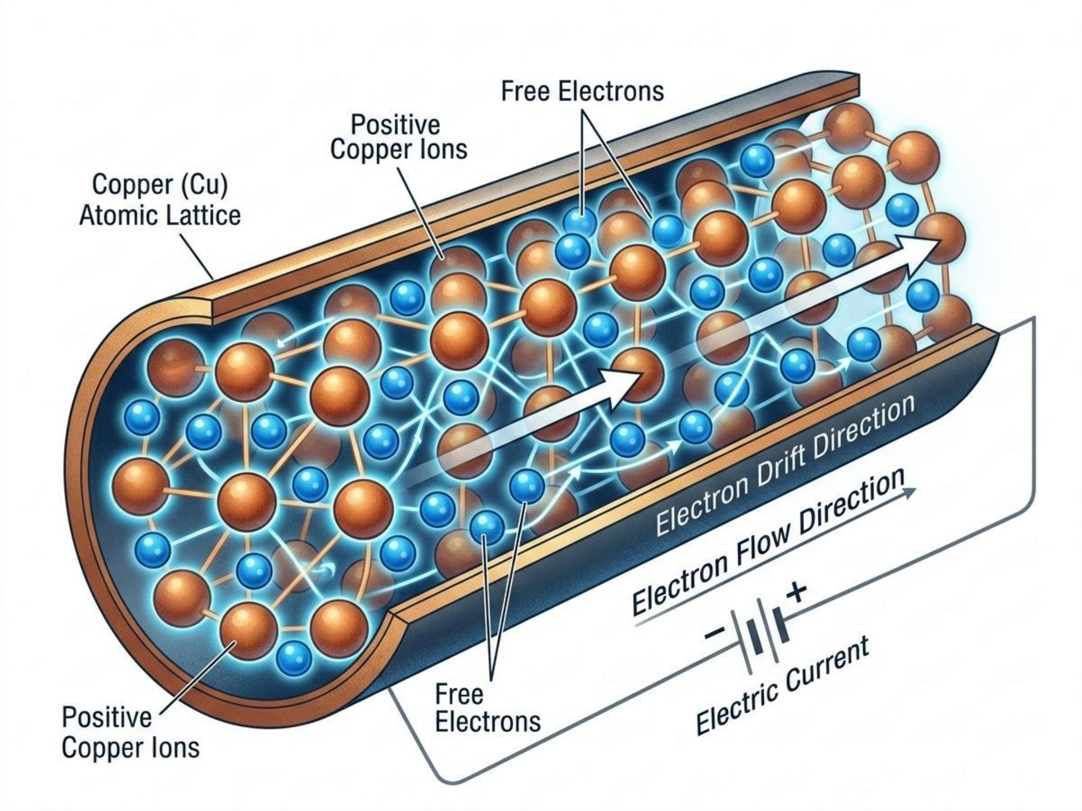


Fig 1.1: Atomic lattice and electron drift representation.

Types of Electricity

Two main categories:

Static Electricity

- Charge stored on the surface of an object.

Current Electricity

- Continuous flow of electric charge.

Static Electricity

Characteristics

Charge is stored on the surface of an object.

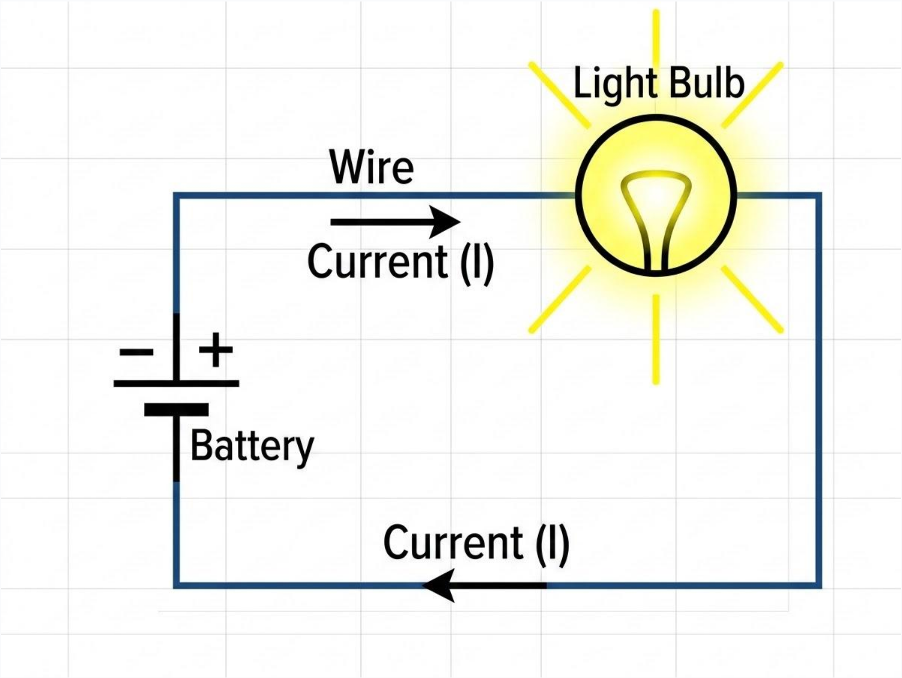
Does not flow continuously.

Examples

Rubbing a balloon on hair

Lightning strikes

Current Electricity



Characteristics

- The continuous flow of electric charge.
- Travels through a conductor (like copper wire).
- Powers electronic devices and circuits.

Types of Current Electricity

Direct Current (DC)

The unidirectional flow of electric charge. The current flows in a constant direction with a steady voltage level.

Commonly Used In:

- Batteries
- Electronic devices
- Solar panels

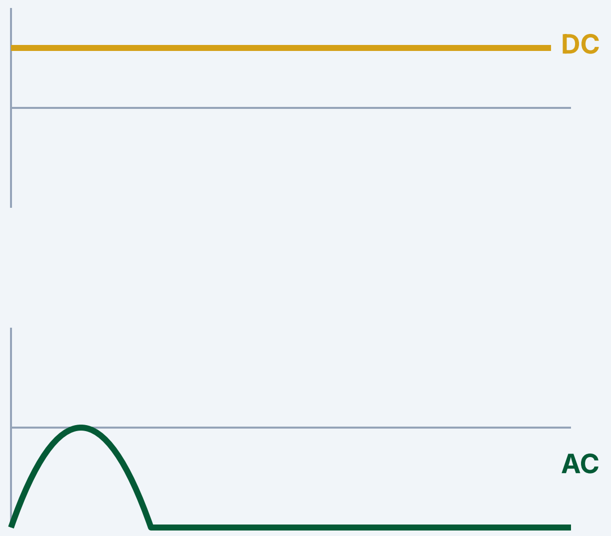
Alternating Current (AC)

An electric current which periodically reverses direction and changes its magnitude continuously with time.

Commonly Used In:

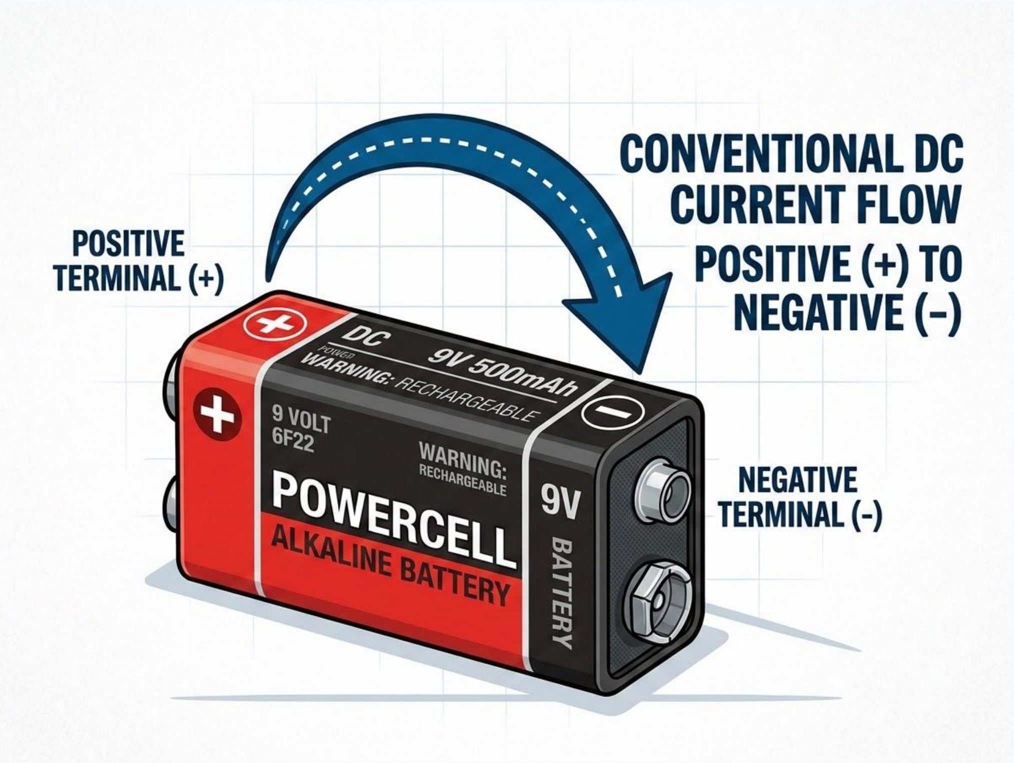
- Home power outlets
- Industrial motors
- Long-distance transmission

Waveform Comparison



Visual representation of voltage over time

Direct Current (DC)

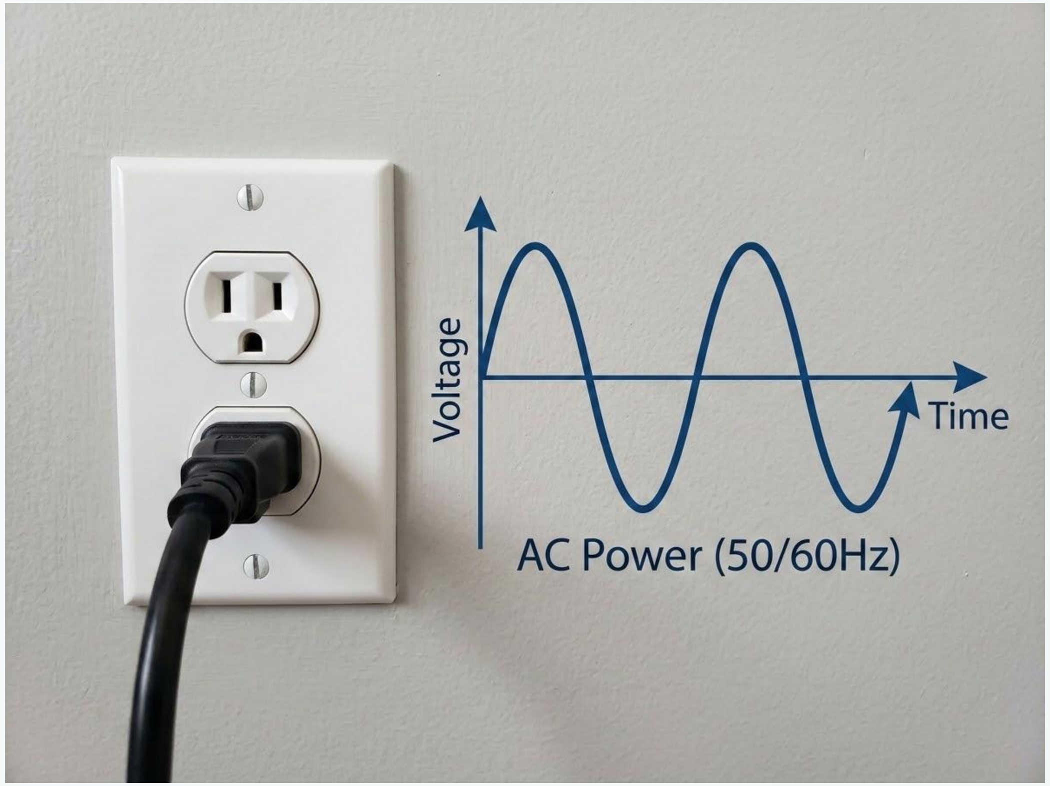


- **Flow:** One direction only.
- **Common Uses:**
  - Batteries
  - Arduino boards
  - Robots
  - DC motors

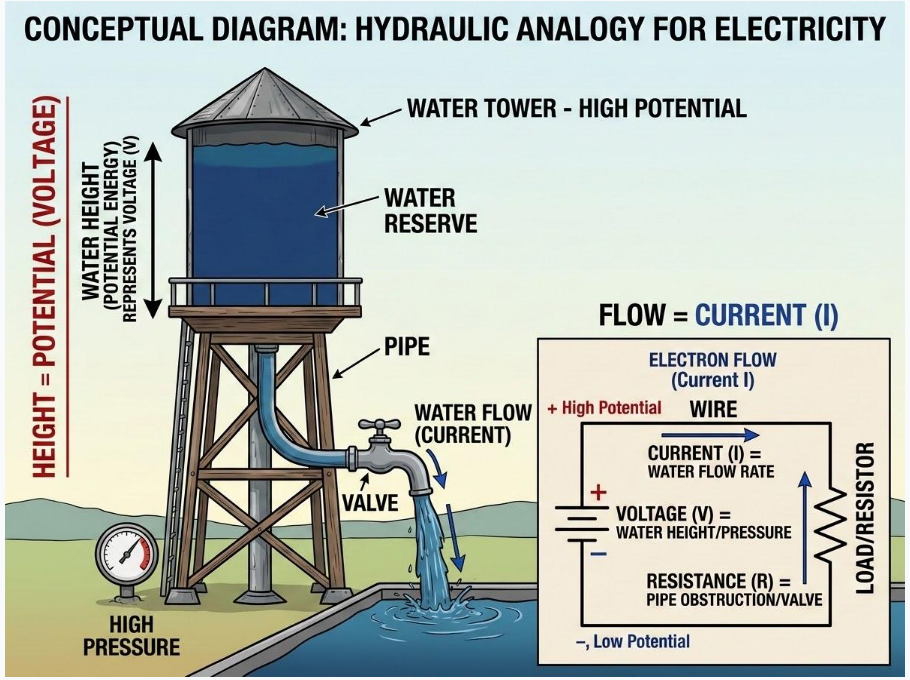
Alternating Current (AC)

- **Flow:** Changes direction repeatedly.
- **Common Uses:**
    - Home electricity (wall outlets)
    - Ceiling fans
    - Household appliances
    - Heavy industrial machines

AC Waveform & Source



Voltage



Definition

The electrical pressure that pushes current through a circuit.

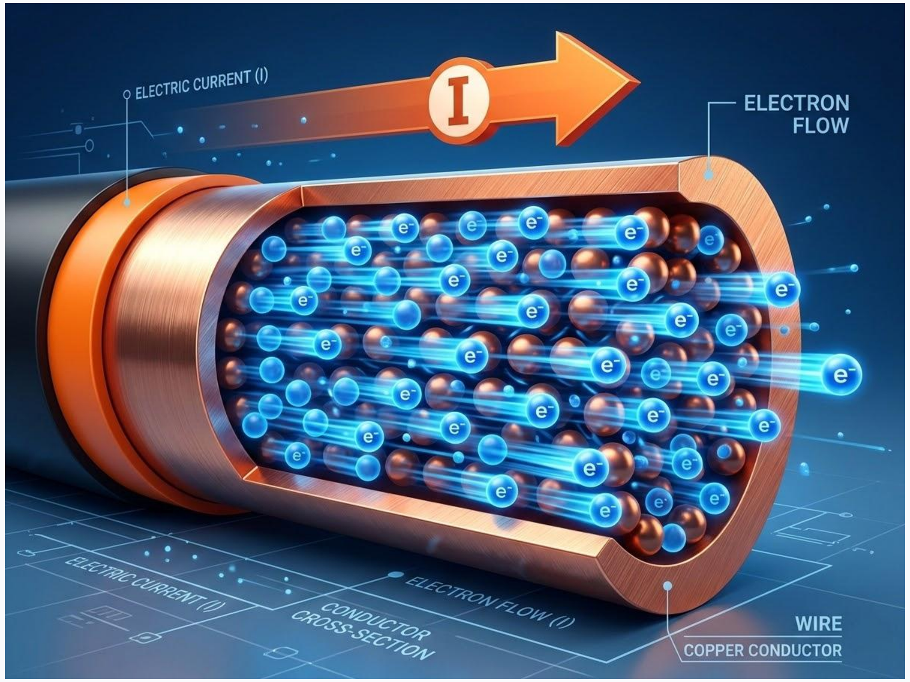
Unit of Measurement

Volt (V)

Analogy

Water pressure in a pipe.

Current



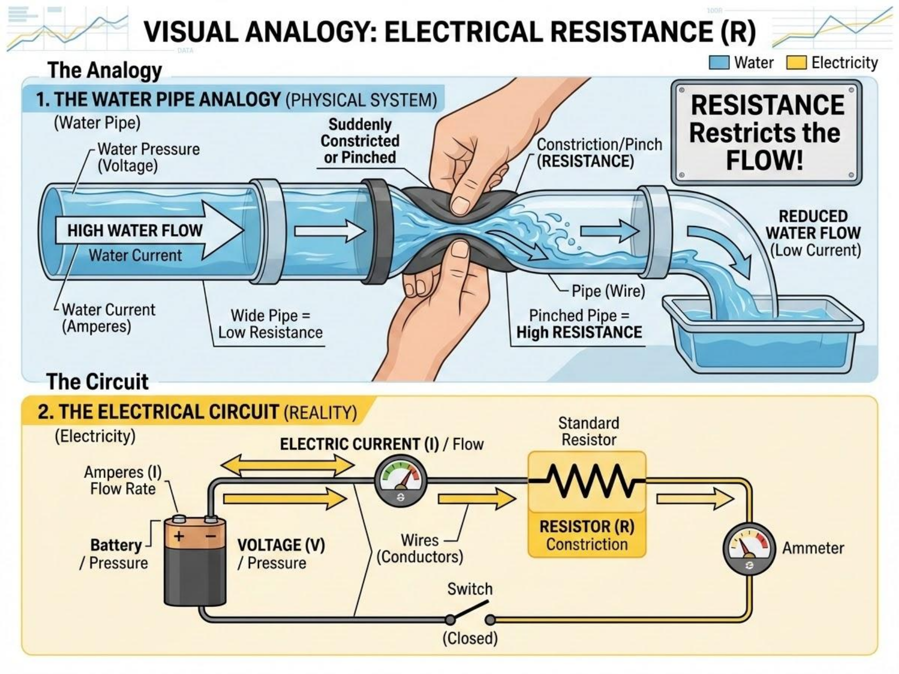
Core Concepts

Definition: The actual flow of electric charge in a circuit.

Unit of Measurement: Ampere (A)

Analogy: The amount of water flowing through a pipe.

Resistance



Definition

The property that opposes the flow of current.

Unit of Measurement

Ohm (Ω)

Analogy

A narrow section or pinch in a water pipe.

Ohm's Law

$$V = I \times R$$

Definitions:

V = Voltage (Measured in **Volts**)

I = Current (Measured in **Amperes**)

R = Resistance (Measured in **Ohms**)

Ohm’s Law Example

Solved Circuit Calculation:

Given

Current (I) = 2A, Resistance (R) = 5Ω

Formula

$V = I \times R$

Calculation

$V = 2 \times 5$

Result

$V = 10V$

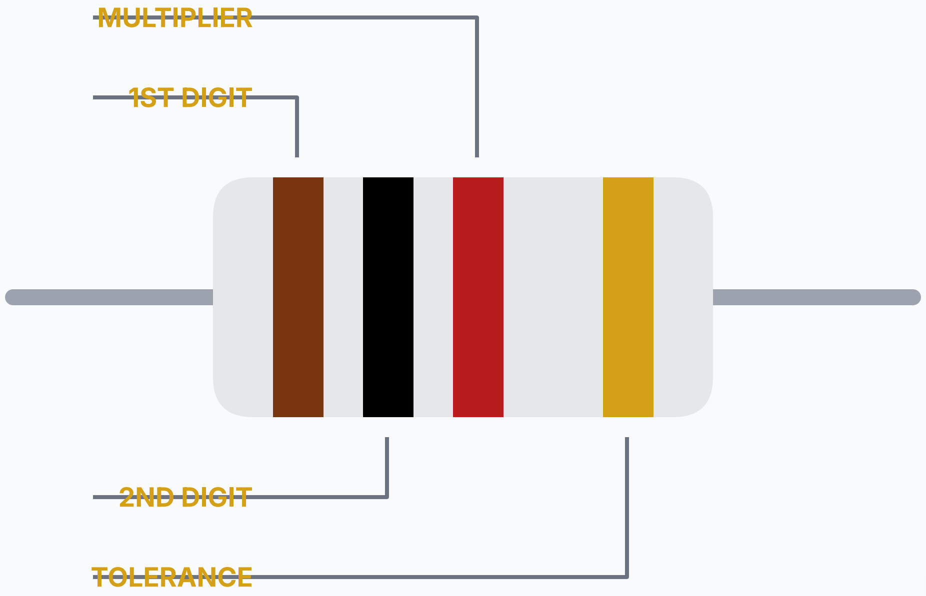
Resistance Color Code Table

Decoding Resistor Values:

Resistors use standard colored bands printed on their bodies to indicate their exact resistance value in Ohms.

Black	0	0	x10^{0} (1Ω)	±20% (None/Blank)
Brown	1	1	x10^{1} (10Ω)	±1% (F)
Red	2	2	x10^{2} (100Ω)	±2% (G)
Orange	3	3	x10^{3} (1kΩ)	
Yellow	4	4	x10^{4} (10kΩ)	
Green	5	5	x10^{5} (100kΩ)	±0.5% (D)
Blue	6	6	x10^{6} (1MΩ)	±0.25% (C)
Violet	7	7	x10^{7} (10MΩ)	±0.1% (B)
Gray	8	8		±0.05% (A)
White	9	9		
Gold	-	-	x10^{-1} (0.1Ω)	±5% (J)

How to Read a Resistor



Band Positions

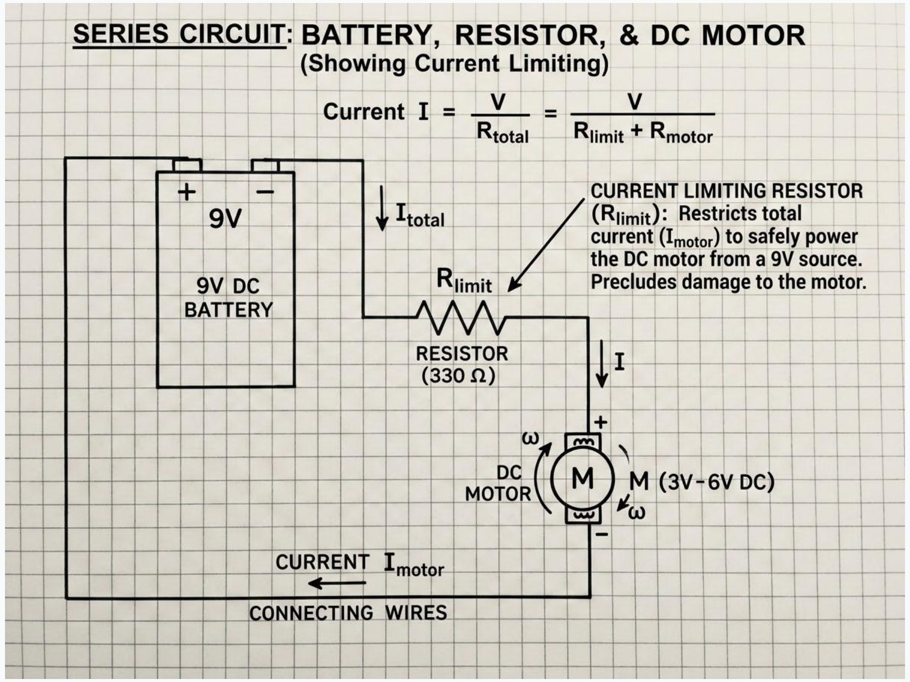
1st Band: First digit of the resistance value.

2nd Band: Second digit of the resistance value.

3rd Band: Multiplier (represents the number of zeros following the first two digits).

4th Band: Tolerance (represents the precision or accuracy percentage).

Finding Resistance



Protecting Components

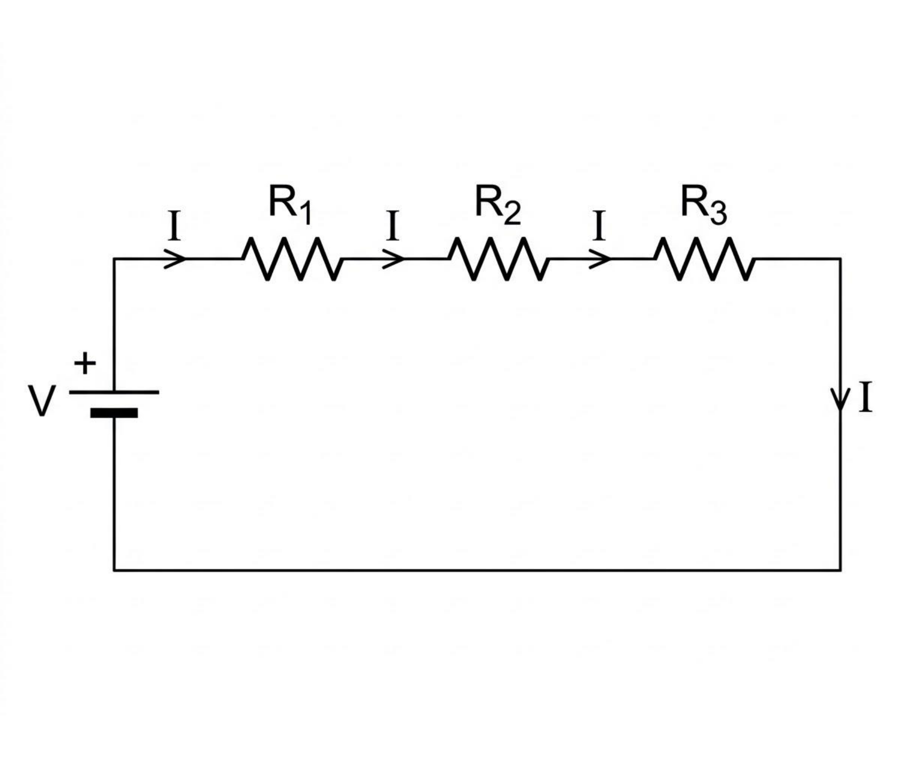
Purpose: Use resistance to limit current and protect sensitive parts (like LEDs or motors).

Ohm's Law Formula

$$R = V / I$$

Where **R** is Resistance (Ohms), **V** is Voltage (Volts), and **I** is Current (Amperes).

Series Resistance



Conceptual Overview

Definition: Resistors connected sequentially, end-to-end.

Current Path: There is only one path for the current to flow.

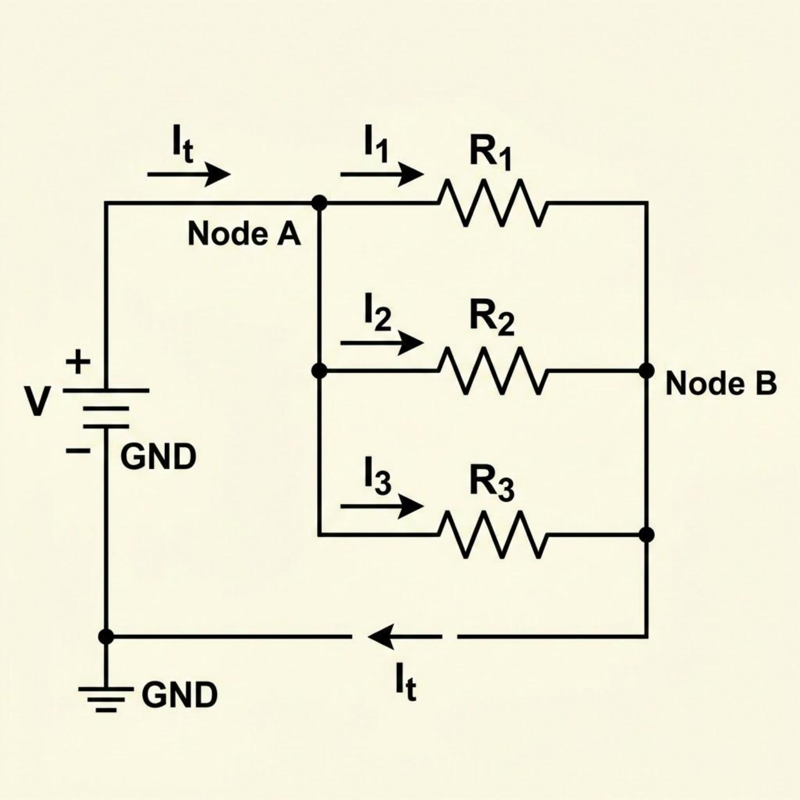
Total Resistance Formula

$$R_{total} = R1 + R2 + R3$$

Parallel Resistance & Module Summary

Parallel Circuits

- Resistors provide multiple paths for current.



Module Wrap-Up

- **Electronics** controls robots.
- **Electricity** powers circuits.
- **Voltage** pushes current.
- **Resistance** controls current.
- **Ohm's Law** calculates circuit values.